Automotive Sales Analytics

Multi-dealer automobile company sells cars from different manufacturers across multiple regions. Management wants to analyse **customer demographics, sales trends, and dealer performance** to improve sales strategy and inventory planning.

Dashboard 1: Executive Sales Overview

Insight 1: Total Revenue and Sales Performance

Finding:

- Total cars sold: 23,906
- Total revenue: $671,525,465
- Average selling price: $28,090 per car
- Analysis period: January 2022–December 2023

Analysis:

The business generated approximately $671.53 million in revenue during the two-year period. The average selling price was approximately $28,090 per vehicle.

This provides a useful baseline for measuring future sales performance and comparing revenue across manufacturers, dealerships, and regions.

Business recommendation: Monitor total sales, revenue, and average selling price monthly to identify changes in business performance.

Insight 2: Sales Increased in 2023

Finding:

Year	Cars Sold	Revenue
2022	10,645	$300.34 million
2023	13,261	$371.19 million
Total	**23,906**	**$671.53 million**

Analysis:

Sales volume increased by approximately **24.57% in 2023** compared with 2022.

Revenue also increased by approximately 23.59%, from $300.34 million to $371.19 million.

This indicates that the business achieved higher sales volume and revenue in 2023. However, the average selling price decreased slightly, from approximately $28,214 in 2022 to $27,991 in 2023.

Business recommendation: Investigate which manufacturers, dealerships, and vehicle categories contributed most to the growth in 2023.

Insight 3: Austin Was the Highest-Selling Region

Finding:

Region	Cars Sold	Revenue
Austin	4,135	$117.19 million
Janesville	3,821	$106.35 million
Scottsdale	3,433	$95.97 million
Pasco	3,131	$88.04 million
Aurora	3,130	$88.69 million
Greenville	3,128	$88.15 million
Middletown	3,128	$87.13 million

Analysis:

Austin was the strongest-performing region, accounting for approximately 17.30% of all recorded sales and 17.45% of total revenue.

Janesville ranked second by sales volume, while Scottsdale ranked third.

The results suggest that Austin is an important market for the business.

Business recommendation: Study Austin's successful sales practices and determine whether similar strategies could improve performance in other regions.

Insight 4: December 2023 Was the Best Revenue Month

Finding:

- Month: December 2023
- Cars sold: 1,921
- Revenue: $54.28 million

December 2023 recorded the highest monthly revenue in the dataset.

November 2023 followed with approximately $51.59 million, while September 2023 generated approximately $51.42 million.

Analysis:

The strong performance in November and December 2023 suggests a potential year-end sales opportunity. However, the available data alone does not establish the cause of the increase.

Business recommendation: Examine year-end promotions, customer demand, inventory availability, and dealership performance to understand the factors behind the higher sales.

Dashboard 2: Customer Behaviour

Insight 5: Male Customers Accounted for Most Purchases

Finding:

Gender	Cars Purchased	Percentage
Male	18,798	78.63%
Female	5,108	21.37%
Total	**23,906**	**100%**

Analysis:

Male customers represented approximately 78.63% of recorded purchases, while female customers represented 21.37%.

The average selling price was approximately $28,039 for male customers and $28,277 for female customers.

The difference in average selling prices was relatively small.

Business recommendation: Study customer preferences, purchasing patterns, and feedback across customer groups to develop more effective marketing strategies. Avoid assuming that gender alone determines purchasing behaviour.

Insight 6: SUVs Were the Most Popular Body Style

Finding:

Body Style	Cars Sold	Percentage of Sales
SUV	6,374	26.66%
Hatchback	6,128	25.63%
Sedan	4,488	18.77%
Passenger	3,945	16.50%
Hardtop	2,971	12.43%
Total	**23,906**	**100%**

Analysis:

SUVs were the most frequently purchased body style, followed closely by hatchbacks.

Together, SUVs and hatchbacks accounted for approximately **52.29% of all sales**.

This suggests that these two categories represent important segments of the recorded market.

Business recommendation: Ensure adequate inventory of SUVs and hatchbacks while continuing to monitor demand for sedans and other vehicle categories.

Insight 7: Customer Income Had Almost No Linear Relationship with Car Price

Finding:

The correlation between annual customer income and car selling price was approximately **0.012**, indicating virtually no linear relationship in this dataset.

Analysis:

Customers with different recorded annual incomes purchased vehicles at broadly similar average prices.

For example, customers with annual incomes between $500,000 and $750,000 had an average car price of approximately $27,763. Customers in the $1 million–$1.5 million income range had an average car price of approximately $28,196.

These findings suggest that annual income alone does not explain the variation in car prices in this dataset.

Business recommendation: Consider additional factors, including vehicle preferences, financing choices, customer requirements, and purchasing motivations, when analyzing customer buying behaviour.

Important: This conclusion concerns the relationship between the two recorded numerical variables. It does not establish that income is unimportant to car-buying decisions in the wider market.

Dashboard 3: Dealer & Product Performance

Insight 8: Automatic Cars Had a Slightly Higher Market Share

Finding:

Transmission	Cars Sold	Percentage
Automatic	12,571	52.59%
Manual	11,335	47.41%
Total	**23,906**	**100%**

Analysis:

Automatic cars accounted for approximately 52.59% of sales, compared with 47.41% for manual cars.

Automatic vehicles therefore had a lead of approximately 5.18 percentage points.

The average selling price was also slightly higher for automatic cars, at approximately $28,249, compared with $27,915 for manual cars.

Business recommendation: Maintain a balanced inventory of both transmission types and evaluate customer preferences by region and vehicle model before making major inventory changes.

Insight 9: Chevrolet Was the Best-Selling Manufacturer

Finding:

The top five manufacturers by sales volume were:

Rank	Manufacturer	Cars Sold	Revenue
1	Chevrolet	1,819	$47.66 million
2	Dodge	1,671	$44.12 million
3	Ford	1,614	$47.23 million
4	Volkswagen	1,333	$34.08 million
5	Mercedes-Benz*	1,285	$34.62 million

The dataset labels the fifth manufacturer as Mercedes-B; the full brand name is shown here for readability.

Analysis:

Chevrolet recorded the highest sales volume, with 1,819 cars sold, representing approximately 7.61% of all recorded sales.

Ford generated approximately $47.23 million in revenue, slightly below Chevrolet's $47.66 million despite selling fewer cars.

This highlights the importance of evaluating both sales volume and revenue when comparing manufacturers.

Business recommendation: Compare manufacturer performance using sales volume, total revenue, average selling price, and demand for individual models rather than relying on sales volume alone.

Insight 10: Progressive Shippers Cooperative Association Led Dealership Sales

Finding:

The leading dealerships by sales volume were:

Rank	Dealership	Cars Sold	Revenue
1	Progressive Shippers Cooperative Association No	1,318	$36.75 million
2	Rabun Used Car Sales	1,313	$37.46 million
3	Race Car Help	1,253	$35.29 million
4	Saab-Belle Dodge	1,251	$35.27 million
5	Tri-State Mack Inc	1,249	$35.09 million

Analysis:

Progressive Shippers Cooperative Association recorded the highest sales volume, with 1,318 cars sold.

However, Rabun Used Car Sales generated higher revenue despite selling five fewer cars. This difference suggests that the average selling price and product mix also influence dealership revenue.

Business recommendation: Compare dealership performance using sales volume, revenue, average selling price, and regional demand. Successful dealership practices could then be evaluated for application elsewhere.

Overall Business Recommendations

Based on the findings across all three dashboards, the following actions are worth considering:

1. **Strengthen high-performing markets:** Investigate the reasons for Austin's strong performance and identify opportunities in other regions.
2. **Plan for seasonal demand:** Examine the factors contributing to the high revenue recorded in November and December 2023.
3. **Focus on popular vehicle categories:** Monitor inventory and demand for SUVs and hatchbacks.
4. **Improve customer segmentation:** Analyze purchasing patterns using multiple customer characteristics rather than relying on gender or income alone.

5. **Optimize vehicle inventory:** Use transmission preferences and model-level sales to guide stock planning.
6. **Evaluate dealership performance comprehensively:** Consider both sales volume and revenue when identifying leading dealers.
7. **Investigate manufacturer performance:** Compare brands by sales, revenue, and average selling price to identify growth opportunities.
8. **Track key performance indicators:** Monitor monthly revenue, sales growth, average selling price, regional performance, and product mix.

Final Conclusion

The Tableau analysis indicates that the business recorded **23,906 car sales and approximately $671.53 million in revenue** between January 2022 and December 2023. Sales volume increased substantially in 2023, Austin was the leading region, SUVs were the most popular body style, and automatic vehicles accounted for a slight majority of sales. Chevrolet led manufacturers by sales volume, while Progressive Shippers Cooperative Association led dealerships.

These insights can help management make more informed decisions about inventory planning, customer segmentation, regional marketing, and dealership performance.

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